2037 - First detection of volatiles from a main-belt comet

Cycle: 1, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

We propose an archival research project to study gas coma volatiles in a main-belt comet with the James Webb Space Telescope's NIRSpec and NIRCam instruments. Main-belt comets have dynamical properties consistent with an origin in the asteroid belt, but exhibit the behavior of comets. Despite repeated mass-loss episodes during perihelion passages consistent with water-ice driven sublimation, volatiles have never been observed at any main-belt comet. Measuring their volatile production rates and coma dust-to-gas mass ratios provides an better understanding of the subsurface preservation of water ice in the asteroid belt. With observations of 238P/Read from GTO program 1252, we expect to exceed previous upper limits by two orders of magnitude, yielding the first detection of water in an MBC coma, or a physically meaningful upper limit.

OBSERVING DESCRIPTION

We do not propose new observations. Instead, observations for this Archival Research project are from JWST program 1252, which targets comet 238P/Read. Comet Read is a main-belt comet, an asteroid with a cometary coma that repeats orbit to orbit. We mean to study the activity of this comet, with NIRSpec spectroscopy and NIRCam imaging. The observations are scheduled for a period when the comet is expected to be active. NIRSpec will observe with the IFU in Prism mode, which covers cometary water and carbon dioxide gas emission bands. NIRCam will observe the comet with two filters, F200W and F277W. F277W includes a water emission band, and F200W will be used to remove the coma continuum from

F277W.